

SERGEY CHERNYAKOV



Senior Backend / Data Engineer · Python · Production Pipelines · AI Integration

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Age: 47 · Male

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Location: Wrocław, Poland · Remote / EU (open to relocation, hybrid 1–2 days/week OK)

PROFESSIONAL SUMMARY

Senior engineer with **20+ years** of commercial experience building **data-driven backend systems, ETL pipelines, integrations and production APIs** for SaaS, fintech, marketing automation, e-commerce, AI and Web3 platforms. Strong hands-on Python and SQL background, with production experience designing and operating data products end-to-end — from ingestion and modelling through analytics and ML-powered automation.

Most recent work centres on **AI / ML engineering, automation pipelines and LLM integrations** in production. Long track record of long-term contracts with international clients in the US, UK, EU and Australia — Top Rated on Upwork, **18,000+ hours, 71+ contracts, 98% Job Success**.

Available as **B2B contractor via Deel.com**.

KEY SKILLS

- **Languages:** Python (primary) · SQL · Ruby · JavaScript / TypeScript · Solidity · (learning Go, Rust)
- **Backend & APIs:** FastAPI · Django · Flask · REST · GraphQL · microservices · event-driven systems · WebSockets · OpenTelemetry tracing
- **Data & ETL:** ETL pipelines · data modelling · batch & streaming · observability and data-quality checks · PostgreSQL · MySQL · ClickHouse · Redis · Kafka · MongoDB · asyncpg
- **Cloud & Infra:** AWS (EC2 / S3 / RDS) · Azure Blob Storage · Docker · Linux · GitHub Actions / GitLab CI · Nginx · Prometheus + Grafana
- **AI / ML in production:** OpenAI API · Anthropic Claude · LLM integration · AI agents · MCP servers · prompt engineering · OCR / document extraction pipelines · RAG · multi-agent systems · XGBoost ML pipelines · schema-versioned datasets
- **Integrations:** Stripe · PayPal · Braintree · Recurly · Bluesnap · YooKassa · Salesforce · Mailchimp · Drip · Aweber · Constant Contact · Twilio · Google Ads / Adwords API · Facebook / Twitter / LinkedIn APIs · blockchain APIs (Binance, Latoken, Cointiger) · SOAP integrations
- **Automation / RPA:** Selenium with Pydantic-validated data models · anti-detection · async PDF downloaders · ZIP batching · Excel reporting
- **Quality:** TDD · RSpec · pytest · production monitoring · performance optimization

PROFESSIONAL EXPERIENCE

AI Engineering Lead / AI Automation Architect — Sabermine (United Kingdom, Remote)

November 2024 – November 2025

Led AI engineering and automation direction for internal and client systems.

- Designed and operated Python-based **ETL and automation pipelines** for data ingestion, transformation and downstream analytics
- Built **multi-agent AI systems** for development, QA, documentation and DevOps automation
- Backend services on **Python, FastAPI, PostgreSQL**; Docker-based deployment, CI/CD via GitHub Actions
- Integrated **OpenAI APIs / LLM workflows** into production for OCR, document classification and content generation
- Built internal tooling to improve data quality, observability and reproducibility of AI pipelines
- Improved development speed and operational efficiency **5–10x** via AI-assisted SDLC

Senior / Principal Backend Engineer — Upwork (International Clients, Remote)

June 2012 – June 2025 (Top Rated)

18,000+ hours · 71+ contracts · 98% Job Success.

 **Selected Python backend systems (2023–2026):**

- **Solana Pulse Bot** (`solana_pulse_bot`) — Real-time data pipeline for Solana memecoin launchpads. Async ingestion via WebSocket + Helius API, asyncpg + psycopg2 dual-pool PostgreSQL, time-aware feature snapshots at T+5/15/30/45/60/75/90/120s checkpoints, ML scoring (XGBoost: 8 heads, 82 features schema v21), deterministic parallelism (state updates sequential in main loop, scoring parallel on frozen snapshots), schema versioning, config-hash drift guards, daily 9 honesty checks (shuffled labels, prior drift, economic backtest), 5-seed feature-stability protocol. **100% backtest-live decision parity** on 300+ tokens.
- **Umetex** (`umetex`) — Django + ChatGPT pipeline for translating medical PDFs/Word/images while preserving complex layout (tables, images, multi-page). Cyrillic font handling, predefined-phrase consistency, IP-based history, real-time progress tracking, asynchronous processing.
- **AQ AI Agent** (`aq_ai_agent`) — FastAPI + Model Context Protocol via Server-Sent Events; exposes legacy Aquarium SOAP CRM as agent-callable MCP tools (legacy SOAP→modern MCP integration pattern).

📁 RPA / data-extraction pipelines:

- **FastDox Exorcist** (`fastdox_exorcist`) — Python pipeline extracting **32,000+ signed CFA documents** from FastDox API v2 (Token auth). Multithreaded downloads of ZIP archives, standardized PDF naming (`req_ID_doc.pdf / req_ID_audit_doc.pdf`), Excel reporting with `client_id / case_ref / journey_id / campaign_id / status / sas_url`, Azure Blob Storage upload with SAS URLs, robust error recovery.
- **SecureMail Automator** (`securemail_automator`) — Barclays SecureMail end-to-end pipeline: CSV/Excel → standardized Excel template → async PDF downloader (retry + logging) → ZIP packaging in user-defined chunks → Selenium-driven SecureMail login + compose + attach + send + delete + detailed CSV reporting.
- **FoS Submission Automator** (`fos_submission_automator`) — Selenium + Pydantic-validated form submission to UK Financial Ombudsman Service; anti-detection (randomized pauses for human-like behaviour); structured client data submission with required + optional fields.
- **Blackhorse DCA Checker** (`blackhorseoffshore`) — Pydantic-modelled form automation for Blackhorse DCA Checker (company/customer/vehicle data validation + Selenium submission).

📁 Backend systems and integrations (long-term):

- **Clickfunnels** (USA, 2012–2017) — Backend engineering on a high-volume marketing-automation SaaS. Built data flows, payment / event pipelines and integrations with **Salesforce, Mailchimp, Drip, Aweber, Constant Contact, GoToWebinar, Google Ads / Adwords API, Stripe, Braintree, Recurly, Twilio**. Stack: Ruby on Rails, MySQL, Kafka, Redis, Docker. Reference: *"Seriously one of the best hires we've ever made on Upwork."* — Ryan Montgomery, COO.
- **Gleam.io** (Australia, 2015–2016) — Backend integrations with social and audience APIs (Facebook, Twitter, Instagram, Pinterest, YouTube, Tumblr, SoundCloud) for SaaS competitions / engagement platform. Stack: Ruby on Rails + Angular, REST APIs, AWS hosting.
- **Lawdingo** (USA, 2010–2012) — Lawyer marketplace with phone-based service and automated payments. Stack: Ruby on Rails, MySQL, AWS (EC2 + Apache+Passenger), Sidekiq, Solr (search), Stripe, Twilio, Capistrano.

📁 Multi-tenant SaaS + enterprise:

- **МаркетПульт** (private, Rails) — multi-tenant SaaS for managing remote outbound sales staff with daily plans, activity proofs, payroll calculation, role-based access.
- **Спец Ресурс** (private, Laravel 12 + React 18 + Inertia.js + PostgreSQL 16) — corporate portal with 5-role access model, mobile UI for field staff, Prometheus + Grafana / Zabbix monitoring.

Senior Fullstack Engineer — Fit2you (Italy, insurance)

April 2022 – January 2024

- Internal enterprise systems for an Italian insurance company (dealer / vendor / client portals)
- Migrated all company applications from **Rails 4.1 → Rails 7.1** (5 legacy apps) with SonarQube quality gates and 80% test coverage
- Supported and developed **all 8 of the company's applications** end-to-end — feature work, maintenance and incident fixes; one of them was a **Node.js + React** app, the rest on Ruby on Rails
- Stack: Ruby, Ruby on Rails, Node.js / Express, TypeScript, Bootstrap, **PostgreSQL**

Senior Ruby on Rails / PHP Engineer — Skykillers Digital Agency (Kyiv, Ukraine)

August 2011 – June 2012

- Lead developer at a digital advertising agency
- Campaign and brand sites for **Colgate, Danone, Nicorette, Visine, Davidoff, Johnson's Baby**, including Facebook campaign apps and promo platforms

Senior PHP Developer — ITC Publishing (Ukrainian media holding)

June 2009 – August 2011

- High-traffic media properties (hotline.ua, itc.ua, dpk.com.ua, video.itc.ua); CMS on Zend Framework, MySQL
- Internal CRM and editorial workflow automation

Earlier — Media Holding RBC-Ukraine, Obozrevatel, Quasar-Micro Techno (KM Core), PR-media

2002 – 2009

Top Ukrainian media portals (rbc.ua, utro.ua, loveplanet.ua, obozrevatel.com), eTutor virtual learning platform. PHP, MySQL, Zend Framework, web analytics, SEO.

SELECTED OPEN-SOURCE / PORTFOLIO PROJECTS

- **solana_pulse_bot** (Python + XGBoost + asyncpg) — Real-time ML pipeline, 100% backtest-live parity. github.com/sergeychernyakov/solana_pulse_bot
- **umetex** (Django + ChatGPT) — Layout-preserving document translation. github.com/sergeychernyakov/umetex
- **file_service** (Flask 3 microservice) — Clean HTTP file-upload service with auth, configurable JSON config, unit tests, healthcheck, minimal deps. github.com/sergeychernyakov/file_service
- **aq_ai_agent** (FastAPI + MCP) — Legacy SOAP CRM exposed as MCP tools. github.com/sergeychernyakov/aq_ai_agent
- **weather_mcp_agent** (FastAPI + MCP SSE) — Clean reference implementation. github.com/sergeychernyakov/weather_mcp_agent
- **blank_python_project** — Python template with Claude Code SDLC agents.

Private: *fastdox_exorcist* (32k+ PDF extraction → Azure Blob), *securemail_automator* (Barclays Selenium pipeline), *fos_submission_automator* (UK FoS form bot), *arbitraflashbot_v2* (DEX arbitrage + Multicall3, 95x perf), *spec_resource* (Laravel 12 + React 18), *marketpult* (multi-tenant Rails SaaS).

EDUCATION

Bachelor's Degree — Don State Technical University, 1999

LANGUAGES

Ukrainian — Native · English — C1 (Fluent professional) · Polish — Learning (A1)

ENGAGEMENT

- **B2B contractor** — invoicing via **Deel.com** (current setup; open to other platforms)
- **Format:** Remote (preferred) · Hybrid OK (1–2 days/week)
- **Location / mobility:** Wrocław, Poland · Open to relocation within EU · Visa sponsorship welcome for non-Poland EU
- **Notice period:** ~1 month
- Strong production / operational mindset; data quality, observability and reproducibility as defaults